Computer Networks

Handwritten Notes for GATE Preparation

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Notes compiled from:
Data Communications and Networking
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Contents

Chapter 1

Physical Layer

1.1 Introduction

To be transmitted, data must be transformed to electromagnetic signals. The Physical layer is concerned with transmitting raw bits over a communication channel. It also deals with framing.

1.1.1 Analog versus Digital Data

- **Analog data** are continuous, while **digital data** have discrete states.
- Analog signals can have an infinite number of values in a range, while digital signals can have only a limited number of values.
- In data communications, we commonly use periodic analog signals (because they need less bandwidth) and non-periodic digital signals (because they can represent variation in data).

1.1.2 Periodic Analog Signals

Periodic analog signals can be classified as simple or composite. A composite analog (periodic) signal is composed of multiple sine waves.

- **Peak amplitude** of a signal is the absolute value of its highest intensity.
- **Period** refers to the amount of time, in seconds, a signal needs to complete one cycle.
- **Frequency** refers to the number of periods in one second.

$$\text{Period} \rightarrow \text{seconds}$$
$$\text{Frequency} \rightarrow \text{Hz} \rightarrow \text{cycles per second}$$
$$f = \frac{1}{T}$$

1.1.3 Phase and Wavelength

- **Phase** describes the position of the waveform relative to time zero.
- **Wavelength** is the distance a single signal can travel in one period.

Wavelength = Propagation Speed × Period

$$\lambda = c \cdot T = \frac{c}{f}$$

1.1.4 Frequency Characteristics

- Frequency is the rate of change with respect to time. Change in a short span of time means high frequency.
- Change over a long span of time means low frequency.
- If a signal does not change at all, its frequency is zero.
- If a signal changes instantaneously, its frequency is infinite.

1.1.5 Single-Frequency Sine Waves

A single-frequency sine wave is not useful in data communications. We need to send a composite signal, a signal made of many simple sine waves.

A normal human being can create a continuous range of frequencies between 0 and 4 kHz.

1.2 Digital Signals

1.2.1 Signal Levels

If a signal has L levels, each level needs $\lceil \log_2 L \rceil$ bits.

$$\text{Number of bits per level} = \lceil \log_2 L \rceil$$

Most digital signals are non-periodic, and thus period and frequency are not appropriate characteristics for digital signals. Bit rate (instead of frequency) is used to describe digital signals.

1.2.2 Bit Rate

Bit rate is the number of bits sent in one second, expressed in bits per second (bps).

Bit Rate = Number of bits sent in 1 sec

1.2.3 Bit Length

The bit length is the distance one bit occupies on the transmission medium.

Bit Length = Propagation Speed × bit duration
= Propagation Speed / Transmission Speed

A digital signal is a composite analog signal with infinite bandwidth.

1.3 Transmission of Digital Signals

1.3.1 Baseband Transmission

Baseband transmission means sending a digital signal over a channel without changing the digital signal to an analog signal.

Baseband transmission requires that we have a low-pass channel, a channel with a bandwidth that starts from zero.

1.3.2 Broadband Transmission

Broadband transmission or modulation means changing the digital signal to an analog signal for transmission.

Modulation allows us to use a bandpass channel – a channel with a bandwidth that does not start from zero.

1.4 Transmission Impairment

$$\text{dB} = 10 \log_{10} \frac{P_2}{P_1}$$

1.4.1 Attenuation

Attenuation means a loss of energy. When a signal travels through a medium, it loses some of its energy in overcoming the resistance of the medium.

To compensate for this loss, amplifiers are used to amplify the signal.

1.4.2 Distortion

Distortion means that the signal changes its shape.

Distortion can occur in a composite signal made of different frequencies because each signal component has its own propagation speed through a medium, and therefore, its own delay in arriving at the final destination.

1.4.3 Noise

Several types of noise, such as thermal noise, induced noise, crosstalk, and impulse noise, may corrupt the signal.

- Induced noise comes from sources such as motors and appliances.
- Crosstalk is the effect of one wire on another.
- Impulse noise is a spike that comes from power lines.

1.5 Signal-to-Noise Ratio (SNR)

$$\text{SNR} = \frac{\text{Average Signal Power}}{\text{Average Noise Power}}$$

- A high SNR means the signal is less corrupted by noise.
- A low SNR means the signal is more corrupted by noise.

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR}$$

1.5.1 Example Question

Theoretically, if we have given a specific bandwidth, we can have any bit rate we want by increasing the number of signal levels.

→ Increasing the levels of a signal may reduce the reliability of the system.

1.6 Data Rate Limits

Data rate depends on three factors:

(1) The bandwidth available

(2) The level of signal we use

(3) The quality of the channel (the level of noise)

1.6.1 Noiseless Channel

Nyquist Bit Rate

$$\textbf{Bit Rate} = 2 \times \text{BW} \times \log_2 L$$

Here, BW is the bandwidth of the channel, L is the number of levels to represent data, and Bit Rate is the bit rate in bits per second.

1.6.2 Noisy Channel

Shannon Capacity

The Shannon capacity gives us the upper limit; it tells us how many signal levels we need.

$$\textbf{Capacity} = \text{BW} \times \log_2(1 + \text{SNR})$$

Capacity $\Rightarrow$ Capacity of the channel in bits per second.

1.6.3 Example Problem

Question: We have a channel with a 1 MHz BW. If the SNR for this channel is 63, what is the appropriate bit rate and signal level?

Solution: Using Shannon formula:

$$\begin{aligned} C &= B \log_2(1 + \text{SNR}) \\ &= 10^6 \log_2(1 + 63) = 10^6 \times 6 \\ &= 6 \text{ Mbps, Ans} \end{aligned}$$

Shannon formula $\Rightarrow$ 6 Mbps; for the upper limit. For better performance, we should choose something lower, 4 Mbps.

Using Nyquist Formula:

$$\begin{aligned} 4 \text{ Mbps} &= 2 \times 1 \text{ MHz} \times \log_2 L \\ &\Rightarrow 2 = \log_2 L \Rightarrow L = 2^2 = 4 \text{ levels} \end{aligned}$$

1.7 Performance of a Network

1.7.1 Bandwidth

1.7.2 Definition

In networking, we use the term **bandwidth** in two contexts:

(i) Bandwidth in hertz refers to the range of frequencies in a composite signal.

(ii) Bandwidth in bits per second refers to the speed of bit transmission in a channel.

Basically, an increase in bandwidth in hertz means an increase in bandwidth in bps.

1.7.3 Throughput

Throughput is a measure of how fast we can actually send data through a network.

Throughput < Bandwidth

The bandwidth is a potential measure of a link; the throughput is an actual measurement of how fast we can send data.

1.7.4 Latency (Delay)

The time taken for an entire message to completely arrive at the destination from the time the first bit is sent out from the source.

Latency = Propagation time + transmission time + queuing time + processing delay

1.7.5 Propagation Time

The time required for a bit to travel from the source to the destination.

Propagation Time $= \frac{\text{Distance}}{\text{Propagation Speed}}$

1.7.6 Transmission Time

Transmission Time $= \frac{\text{Message Size}}{\text{Bandwidth}}$

1.7.7 Queuing Time

The time needed for each intermediate or end device to hold the message before it can be processed.

1.7.8 Bandwidth-Delay Product

It is the number of bits that can fill the link.

Bandwidth-Delay Product = Bandwidth × Delay

1.7.9 Jitter

Jitter is an problem if different packets of data encounter different delays and the application using the data at the receiver site is time-sensitive (audio and video data).

1.7.10 Example Problem

Question: What are the propagation time and the transmission time for a 5 MB message if the bandwidth of the network is 1 Mbps? Assume that the distance between the sender and the receiver is 12,000 km and that light travels at 2.4×10^8 m/s.

Solution:

$$\begin{aligned}\text{Propagation Time} &= \frac{12,000 \times 10^3}{2.4 \times 10^8} \\ &= 5 \times 10^{-2} \text{ sec} \\ &= 0.05 \text{ sec}\end{aligned}$$

$$\begin{aligned}\text{Transmission Time} &= \frac{5 \times 10^6 \times 8}{1 \times 10^6} \\ &= 40 \text{ sec}\end{aligned}$$

$$\begin{aligned}\text{Latency} &= \text{Propagation time} + \text{transmission time} \\ &= 40 + 0.05 \\ &= 40.05 \text{ sec} \approx 40 \text{ sec}\end{aligned}$$

1.8 Signal Elements Versus Data Elements

1.8.1 Definitions

- A **data element** is the smallest entity that can represent a piece of information; this is the bit.
- In data communications, a **signal element** carries data elements. A signal element is the shortest unit of a digital signal.

We define a ratio r, which is the number of data elements carried by each signal element.

1.8.2 Data Rate Versus Signal Rate

1.8.3 Definitions

- The **data rate** defines the number of data elements (bits) sent in one second. The unit is bits per second (bps).
- The **signal rate** is the number of signal elements sent in one second. The unit is the baud.

1.8.4 Formulas

For digital signals:

$$\boxed{S = c \times N \times \frac{1}{r} \text{ baud}}$$

For analog signals:

$$\boxed{S = N \times \frac{1}{r} \text{ baud}}$$

where S = Signal Rate (baud), c = case factor, N = data rate (bps), and r = number of data elements per signal element.

1.9 Asynchronous Transmission

In asynchronous transmission, the timing of a signal is not important. Instead, information is received and translated by agreed-upon patterns.

In asynchronous transmission, we send one start bit (0) at the beginning and one or more stop bits (1s) at the end of each byte. There may be a gap between each byte.

1.9.1 Synchronous Transmission

In synchronous transmission, we send bits one after another without start or stop bits or gaps. It is the responsibility of the receiver to group the bits.

Synchronous transmission is faster than asynchronous transmission.

1.10 Transmission Media

Transmission media can be divided into two broad categories:

(i) Guided Media

(ii) Unguided Media

1.10.1 Guided Media

Guided media include twisted pair, coaxial cable, and fiber-optic cable.

1.10.2 Twisted Pair Cable

→ RJ45 connector is used.

→ Used in telephone lines, DSL lines, and LANs.

1.10.3 Coaxial Cable

→ Carries signals of higher frequency ranges than those in twisted-pair cable.

→ Higher frequency means higher bandwidth.

→ Used in Cable TV, Thin Ethernet, and Thick Ethernet.

→ Attenuation is much higher in coaxial cable than in a twisted-pair cable.

→ Coaxial cable has much higher bandwidth.

→ It requires the frequent use of repeaters.

1.10.4 Unguided Media (Free Space)

1.10.5 Types

Radio Waves

→ Radio waves have frequencies between 3 kHz and 1 GHz.

→ Radio waves are omnidirectional.

→ Used for long-distance broadcasting.

→ Can penetrate walls.

→ Have low data rate for digital communication.

Microwaves

→ Have frequencies between 1 and 300 GHz.

→ Microwaves are unidirectional.

→ Very high-frequency microwaves cannot penetrate walls.

→ High data rate for digital communication is possible.

→ Used for unicast (one-to-one) communication such as cellular phones, satellite networks, and wireless LANs.

Infrared

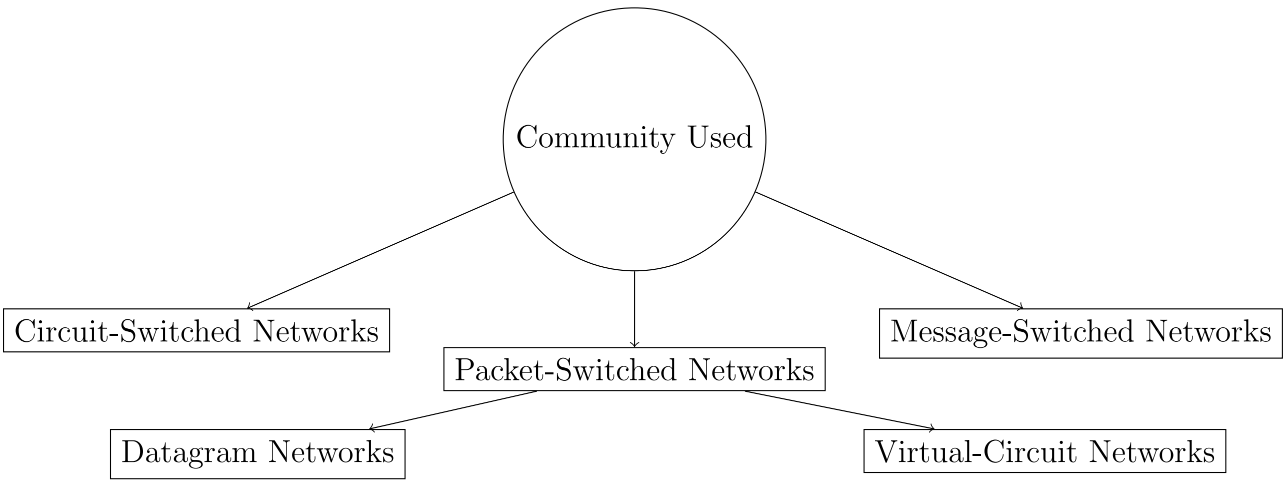
→ Have frequencies between 300 GHz and 400 THz.

→ Used for short-range communication.

→ Cannot penetrate walls.

1.11 Switching

A switched network consists of a series of interlinked nodes, called switches.



1.11.1 Circuit-Switched Networks

It is made of a set of switches connected by physical links, in which each link is divided into n channels.

Circuit switching takes place at the physical layer.

The resources need to be reserved during the setup phase and remain dedicated for the entire duration of data transfer until the teardown phase.

There is no addressing involved during data transfer.

1.11.2 Datagram Networks

In a packet-switched network, there is no resource reservation; resources are allocated on demand.

Message is divided into packets of fixed or variable size.

In a datagram network, each packet is treated independently of all others.

Datagram switching is normally done at the network layer.

The switches in a datagram network are referred to as routers.

Datagram networks are sometimes referred to as connectionless networks.

1.11.3 Virtual Circuit Networks

A virtual-circuit network is a cross between a circuit-switched network and a datagram network.

All packets follow the same path established during the connection.

It is implemented in the data link layer.

Chapter 2

Network Topologies

2.1 Bus Topology

→ Single cable connects all computers.

→ Each computer has connector to shared cable.

→ Computers must synchronize and allow only one computer to transmit at a time.

→ Cable segment must end with a terminator.

→ Uses thin coaxial cable (backbones will be thick coaxial cable).

→ Standard is IEEE 802.3.

2.1.1 Advantages

(i) Inexpensive to install

(ii) Easy to add stations

(iii) Uses less cable than other topologies

(iv) Works well for small networks

2.1.2 Disadvantages

(i) No longer recommended

(ii) Backbone breaks, while network is down

(iii) Limited number of devices can be attached

(iv) Difficult to isolate problems

(v) Sharing same cable slows response rate

2.2 Star Topology

→ Each device has a dedicated point-to-point link to a hub.

→ Easier to maintain.

→ Very flexible and the most preferred topology by industries.

→ Requires more cable than bus topology.

→ If hub goes down, the whole system is dead.

→ Used in high-speed LANs.

→ Twisted pair cable is used.

The hub can be used in two ways:

(i) As a repeater, where a frame from one node is retransmitted to all of the outgoing links.

(ii) As a switch, where a frame from one node is retransmitted only on an outgoing link to the destination station.

2.3 Ring Topology

→ Computers connected in a closed loop.

→ Most common type is Token Ring (IEEE 802.5).

→ Single ring: Data travels in one direction.

→ Double ring: Allows fault tolerance.

2.3.1 Advantages

→ Data packets travel at great speed

→ No collisions

→ Easier to find faults

→ No terminators required

2.3.2 Disadvantages

→ A break in the ring will bring it down

→ Not as common as the bus or star devices available

Widely used in Wide Area Networks.

2.4 Mesh Topology

Not common on LANs. Most often used in WANs to interconnect LANs.

Each node is connected to every other node.

It is fault-tolerant.

2.4.1 Advantages

→ Improved fault tolerance

→ Can carry more data

2.4.2 Disadvantages

→ Expensive

→ Difficult to install, manage, and troubleshoot

2.5 Physical versus Logical Topology

The actual layout of a network and its media is its physical topology.

The way in which the data accesses the medium and transmits packets is the logical topology.

A glance at a network is not always revealing. Cables emerging from a hub does not make it necessarily a star topology – it may actually be a bus or a ring.

2.6 Combination of Topology and Transmission Media

→ Twisted pair is suitable for use in star and ring topologies.

→ Coaxial cable is suitable for use in bus topology.

→ Fiber optics is suitable for use in ring and star topology.

→ Unguided media are suitable for star topology.

Chapter 3

ISO/OSI Stack

3.1 Protocol Hierarchy

Most networks are organized as a series of layers.

The task of each layer is to give some services to the upper layer. Each layer maintains a virtual connection with the corresponding layers in a peer.

The interface between the layers in the same node is well defined.

The implementation of each layer in each node is transparent to other nodes.

The protocols between the peer layers can be changed if the peers all agree. However, it need not be referred to other layers.

3.2 Service Definitions

The service definitions tell what the layer does and nothing else.

The interface tells the process above it how to access it. It specifies what the parameters are and what results to expect.

3.3 Open System Interconnection Reference Model (OSI-ISO)

The OSI reference model is a conceptual model composed of seven layers, each specifying particular network functions.

It was developed by the International Organization for Standardization (ISO).

The OSI model is not a protocol.

APPLICATION	7
PRESENTATION	6
SESSION	5
TRANSPORT	4
NETWORK	3
DATA-LINK	2
PHYSICAL	1

3.3.1 Upper Layers of the OSI Model

The upper layers of the OSI model generally are implemented only in software.

The physical layer and the data link layer are implemented in software and hardware.

3.4 A Given Layer in the OSI Model

A given layer in the OSI model generally communicates with three other OSI layers:

(i) The layer directly above it

(ii) The layer directly below it

(iii) Its peer layer in other networked systems

OSI is a model, not a protocol.

The OSI model provides a conceptual framework between computers, but the model itself is not a method of communication. Actual communication is made possible by using communication protocols.

A protocol implements the functions of one or more of the OSI layers.

3.5 Functions of the OSI Layers

3.5.1 Physical Layer

The physical layer is concerned with transmitting raw bits over a communication channel.

The physical layer is also concerned with framing.

Physical Characteristics of Interfaces and Medium

The number of pins and functions of each pin of the network connector (Mechanical).

Representation of Bits

How 0s and 1s are changed to electrical signals.

Data Rate and Synchronization of Bits

Sender and receiver must use the same bit rate and both must have synchronized clocks.

Establishing and Breaking of Connection

Transmission Mode

Simplex, half-duplex, or full-duplex.

Physical Topology

Bus/Star/Ring/Mesh.

3.5.2 Data Link Layer

The data link layer transforms the physical layer, a raw transmission facility, to a reliable link. It makes the physical layer appear error-free to the upper layer (network layer).

Framing

The data link layer divides the stream of bits received from the network layer into manageable data units called frames.

Physical Addressing

The data link layer adds a header to the frame to define the address of the next hop in the route. Next hop (node) may be the receiver or some intermediate node.

Flow Control

Deals with the data rate mismatch between the sender and the receiver.

Error Control

Uses mechanisms to detect and retransmit damaged or lost frames, and to recognize duplicate frames by using error control ARQ technique.

Access Control

Uses protocols like ALOHA and CSMA to control access on a shared medium.

Synchronization

Synchronize and initialize send and receive sequence numbers with its peer at the other end.

3.5.3 Network Layer

The network layer is responsible for the source-to-destination delivery of a packet.

If two systems are connected to the same link, there is usually no need for a network layer.

This source-to-destination delivery is performed using two basic approaches known as connection-oriented or connectionless network layer services.

Logical Addressing

The network layer adds a packet header to the frame coming from the upper layer that, among other things, includes the logical addresses of the sender and receiver.

Routing

The routing algorithm is the piece of software that decides where a packet goes next.

Dealing with Congestion

Congestion occurs in a network when more packets enter an area than can be processed.

Dealing with Internetworking

Packets may travel through many different networks. Each network may have a different frame format.

Some networks may be connectionless, others connection-oriented.

Source-to-destination delivery is also called as end-to-end delivery or host-to-host delivery.

3.5.4 Transport Layer

The transport layer is responsible for process-to-process delivery of the entire message.

The transport layer ensures that the whole message arrives intact and in order, overseeing both error control and flow control at the source-to-destination level.

Service Point Addressing

The transport layer header includes a type of address called a Service Point Address or Port Address to deliver the entire message to the correct process on the computer.

Segmentation and Reassembly

A message is divided into transmittable segments with each segment containing a sequence number, that is used for reassembly at the message correctly.

Connection Control

The transport layer can be either connectionless or connection-oriented. A connectionless transport layer treats each segment as an independent packet. A connection-oriented transport layer makes a connection with the transport layer at the destination machine first before delivering the packets.

Flow Control

Flow control at this layer is performed end-to-end.

Error Control

Error control at this layer is performed process to process. Error correction is usually achieved through retransmission.

3.5.5 Session Layer

This layer allows users on different machines to establish sessions between them. A session allows ordinary data transport but it also provides enhanced services useful in some applications.

Some of the session-related services are:

(i) **Dialogue Control:** Session can allow traffic to go in both directions at the same time, or in only one direction at one time. It also decides who speaks when and how long.

(ii) **Token Management:** Allows only one side that is holding token to perform the critical operation.

(iii) **Synchronization:** Allows a process to add checkpoints, or synchronization points in a stream of data, so that after a crash, only the data transferred after the last checkpoint have to be repeated.

Session layer establishes, maintains, and synchronizes the interaction among communicating systems.

3.5.6 Presentation Layer

This layer is concerned with syntax and semantics of the information transmitted, unlike other layers which are interested in moving data reliably from one machine to other.

Few of the services that presentation layer provides are:

(i) **Translation:** Encoding data in a standard agreed-upon way.

(ii) **Encryption and Decryption**

(iii) **Compression and Decompression**

3.5.7 Application Layer

The application layer enables the user to access the network.

Various services provided by application layer are:

(i) File Transfer (FTP)

(ii) Remote Login (Telnet)

(iii) Mail (SMTP)

(iv) News (NNTP)

(v) Web (HTTP)

Chapter 4

Data Link Layer

Specific responsibilities of the data link layer include framing, addressing, flow control, error control, and media access control.

4.1 Error Detection and Correction

4.1.1 Types of Errors

(i) Single-bit error

(ii) Burst error

If the length of the burst is measured from the first corrupted bit to the last corrupted bit, the burst length would be determined.

4.1.2 Redundancy

To detect or correct errors, we need to send extra (redundant) bits with data.

4.1.3 Forward Error Correction

The receiver tries to guess the message by using redundant bits.
In backward error correction, we retransmit the frame.

Modulo-2 Arithmetic

$$\begin{aligned}
&0+0=0, \quad 0+1=1\\
&0-0=0, \quad 0-1=1\\
&1+0=1, \quad 1-0=1\\
&1-1=0
\end{aligned}$$

4.2 Block Coding

In block coding, we divide our message into blocks, each of k bits, called datawords. We add r redundant bits to each block to make the length $n = k + r$. The resulting n-bit blocks are called codewords.

4.2.1 Hamming Distance

The Hamming distance between two words is the number of differences between corresponding bits.

$$d(10101, 11110) = 3$$

4.2.2 Minimum Hamming Distance

The smallest Hamming distance between all possible pairs in a set of words.

4.2.3 Minimum Distance for Error Detection

To guarantee the detection of up to s errors in all cases, the minimum Hamming distance in a block code must be $d_{\text{min}} = s + 1$.

4.3 Minimum Distance for Error Correction

To guarantee correction of up to t errors in all cases, the minimum Hamming distance in a block code must be more than two.

Number of s.

If R is the number of additional bits, the condition is:

$$2^R \geq m + R + 1$$

Number of Data Bits (m)	**Number of Redundancy Bits (R)**	**Total Bits (m+R)**
1	2	3
2	3	5
3	3	6
4	3	7
5	4	9
6	4	10
7	4	11
8	4	12

4.4 Error Detection Techniques

4.4.1 Simple Parity Check

Append a parity bit to the end of the data.

There are two types of parity: even parity and odd parity.

It can detect all single-bit errors.

It can also detect burst errors, if the number of bits in error is odd.

4.4.2 Two-Dimensional Parity Check

Organizes the block of bits in the form of a table.

Parity check bits are calculated for each row, and after that for all columns.

It can detect up to three errors that occur anywhere in the table. However, errors affecting 4 bits may not be detected.

4.4.3 Checksum

The Sender's End

The data is divided into k segments each of m bits. The segments are added using 1's complement to get the sum.

The sum is complemented to get the checksum.

The checksum segment is sent along with data segments.

The Receiver's End

All received segments are added using 1's complement arithmetic to get the sum.

The sum is complemented.

If the result is zero, the received data is accepted; otherwise discarded.

Example 4.4.1. Sender:

$$\begin{aligned}
E &= 4, m = 8 \\
(i) &\rightarrow 10110011 \\
(ii) &\rightarrow 10101010 \\
&\quad\; 01011110 \\
&\quad\; C_1C_2C_3C_4C_5C_6 \\
\text{Sum} &: 00101110
\end{aligned}$$

Receiver:

$$\begin{aligned}
(i) &\rightarrow 10110011 \\
(ii) &\rightarrow 10101010 \\
(iii) &\rightarrow 01011110 \\
&\quad\; 00101110 \\
&\quad\; 10100101 \\
\text{Sum} &: 11111111 \\
\text{Complement} &: 00000000 \\
\text{Conclusion} &: \text{Accept data}
\end{aligned}$$

4.5 Performance of Checksum

→ CRC can detect all single-bit errors.

→ CRC can detect all double-bit errors (provided that the minimum three terms in the characteristic polynomial $P(x)$).

→ CRC can detect all odd-number of errors (provided that $P(x)$ should be divisible by $c + 1$).

→ CRC can detect all burst errors of length less than the degree of the characteristic polynomial (n).

→ CRC detects most of the larger burst errors with a high probability.

4.6 Cyclic Redundancy Check (CRC)

4.6.1 Basic Approach

Given an m-bit block of data, the sender generates an n-bit sequence, known as a Frame Check Sequence (FCS), so that the resulting frame consists of $m + n$ bits.

The receiver divides the incoming frame by the number used for generating FCS. If there is no remainder, the receiver assumes that there is no error.

4.6.2 Generating FCS

To generate an n-bit FCS, the following steps are followed:

(i) Append n 0s to the right-end side of the m-bit dataword.

(ii) The $(m + n)$-bit result is fed into the generator.

(iii) The generator divides the augmented dataword by the divisor (modulo-2 division), of size $n + 1$, that is predefined and agreed-upon.

(iv) The n-bit remainder of the division is called the frame check sequence (FCS).

(v) The FCS is appended to the dataword to create a codeword.

4.6.3 Example

Data: 1010, Divisor: 1011

At sender:

$$
\begin{array}{l}
1001 \rightarrow \text{Quotient} \\
1011 \mid 1010000 \\
\qquad 1010 \\
\qquad \underline{1011} \\
\qquad 0010 \\
\qquad \underline{0000} \\
\qquad 1000 \\
\qquad \underline{1011} \\
\qquad 011 \rightarrow \text{Remainder}
\end{array}
$$

Data to be sent: 1010011
At receiver:

Since remainder is 0, there is no error

4.7 Performance of CRC

→ CRC can detect all single-bit errors

→ CRC can detect all double-bit errors

→ CRC can detect all odd number of errors

→ CRC can detect all burst errors

→ CRC detects most of the larger burst errors with a high probability

4.8 Error Correction Technique

4.8.1 Requirement for Error Detection

The minimum Hamming distance between any two codewords should be two.

4.8.2 Requirement for Error Correction

The minimum Hamming distance between any two codewords must be more than two.

Number of additional bits should be such that it can point to the position of the bit in error.

If R is the number of additional bits, the condition is:

$$2^R \geq m + R + 1$$

4.9 Hamming Code

4.9.1 Example

Data ⇒ 1011

Hamming Code ⇒

7	6	5	4	3	2	1
d_4	d_3	d_2	p_3	d_1	p_2	p_1

Codeword ⇒ 1010101

To each group of m information bits, R parity bits are added to form $(m + R)$ bit codeword.

Location of each of the $(m + R)$ bits is assigned a decimal value.

The R parity bits are placed in positions $1, 2, 4, \ldots, 2^{R-1}$.

R parity checks are performed on selected digits of each codeword.

At the receiving end, the parity bits are recalculated. The decimal value of R parity bits provides the bit position in error, if any.

4.9.2 Calculating Parity Bits

$$P_1 \to 1, 3, 5, 7$$
$$P_2 \to 2, 3, 6, 7$$
$$P_3 \to 4, 5, 6, 7$$

4.9.3 Example Problem

Data $\Rightarrow$ 1011

5		4		3		2	1
d_4	d_3	d_2	p_3	d_1	p_2	p_1	
1	0	1	0	1	0	1	

Sent to Receiver $\to$

Received Codeword: 1 0 1 0 1 0 1

Error

Calculating parity bits:

$$\begin{array}{ccc} C_3 & C_2 & C_1 \\ 1 & 0 & 1 = 5 \end{array}$$

Location of error

After error correction:

$$1 \quad 0 \quad 1 \quad 1 \quad 0 \quad 1$$

Data to be sent: $x^4 + x^3 + x + 1$

CRC:

$$\frac{x^6 + x^4 + x^3 + x + 1}{\text{CRC}}$$

At Receiver:

$$x^4 + x^4 + x + 1 \to M(x) = x^3(x + 1)$$
$$\text{Divisor} = 1011 = x^3 + x + 1$$

4.10 Flow Control

Flow control is the set of procedures used to restrict the amount of data that the sender can send before waiting for acknowledgment.

Various protocols used in flow control are:

(1) Stop-and-Wait Flow Control

(2) Sliding Window Flow Control

4.10.1 Stop-and-Wait Flow Control

This is the simplest form of flow control.

The source transmits a data frame.

After receiving the frame, the destination indicates its willingness to accept another frame by sending back an ACK frame acknowledging the frame just received.

The source must wait until it receives the ACK frame before sending the next data frame.

With the use of multiple frames for a single message, the stop-and-wait protocol does not perform well because only one frame can be in transit at a time.

4.10.2 Sliding Window Flow Control

In sliding window protocol, we allow multiple frames to be in transit at the same time.

To keep track of the frames, sender sends sequentially numbered frames.

The sequence numbers range from 0 to $2^k - 1$.

The sender maintains a list of sequence numbers that it is allowed to send (sender window).

The size of the sender window is at most 2^{k-1}.

For $R = 3$,

$$0 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6 \quad 7 \quad 0$$

The receiver also maintains a window of size 1.

The receiver acknowledges a frame by sending an ACK frame that includes the sequence number of the next frame expected.

This scheme can be used to actually acknowledge multiple frames.

4.10.3 Piggybacking

If two stations exchange data, each need to maintain two windows: one for transmit and one for receive. To save communication capacity, a technique called piggybacking is used.

When a frame is carrying data from A to B, it can also carry control information about arrived (or lost) frames from B and vice-versa.

If a station has an ACK but no data to send, it sends a separate ACK frame.

4.10.4 Performance of Flow Control Protocols

4.10.5 Stop and Wait Protocol

Utilization or Efficiency of the Line

$$U = \frac{\text{transmission}}{2 \times \text{propagation} + \text{transmission}}$$

Assuming transmission time of ACK to be negligible ($\text{tACK} = 0$):

$$U = \frac{\text{transmission}}{2 \times \text{propagation} + \text{transmission}}$$

Sliding Window Protocol

$$U = \frac{W \times \text{transmission}}{2 \times \text{propagation} + \text{transmission}}$$

where W = Window size (2^{k-1}), propagation = Propagation time, transmission = Transmission time.

For maximum utilization, $U = 1$:

$$W \times \text{transmission} \geq 2 \times \text{propagation} + \text{transmission}$$

then $U = 1$.

4.11 Backward Error Control

During frame transmission, following two types of errors may occur:

4.11.1 Types of Errors

Lost Frame

A noise burst may damage a frame to such an extent that it is not recognizable at the receiving end.

Damaged Frame

A frame does arrive, but some of the bits are in error.

Error control in the data link layer is based on automatic repeat request (ARQ), which is the retransmission of data.

4.11.2 Three Versions of ARQ

(i) Stop-and-Wait ARQ

(ii) Go-back-N ARQ

(iii) Selective Repeat ARQ

4.11.3 Stop-and-Wait ARQ

Based on the stop-and-wait flow control technique.

The source transmits a single frame and then waits for an ACK.

No other data frame can be sent until the ACK arrives at the source.

A timer is used to take care of lost and damaged frames and lost ACK.

The sender will timeout and resend the same frame, if the ACK is lost.

The receiver receives the same frame twice. How to identify duplicate frames?

A 1-modulo-2 numbering scheme is used, where frames are alternately labeled with 0 or 1, and positive acknowledgments are of the form ACK0 or ACK1.

The ACK always announces the sequence number of the next frame accepted.

4.11.4 Go-back-N ARQ

Based on sliding window protocol and most widely used.

The source station may send a series of frames sequentially up to a maximum number, without hearing about any ACK.

If no error, ACK as usual.

In case of no error, the destination will acknowledge incoming frames as usual.

If the destination detects error in frame, or it received a frame out of order (frame lost), it sends a NAK for that frame (using reject or REJ frame).

The destination will discard that frame and all future frames until the frame in error is correctly received.

The source, on receiving a REJ, must retransmit the frame in error plus all succeeding frames.

For lost ACK, the sender after timeout resends the frames that are still unacknowledged.

For 8-bit sequence number, the maximum sender window size is limited to $(2^k - 1)$.

The size of the receiver window is 1.

4.11.5 Selective Repeat ARQ

In this case, only those frames are retransmitted for which negative ACK has been received (called selective reject SREJ) or time-out has occurred.

Receiver requires storage buffers to contain out-of-order frames until the frame in error is correctly received.

The sender and receiver window must be at most one-half of 2^k:

$$\Rightarrow \frac{2^k}{2} = 2^{k-1}$$

4.12 Data Link Layer Protocols

4.13 HDLC Protocol

HDLC

→ High-Level Data Link Control

→ Bit-oriented protocol

→ Communication over point-to-point and multipoint links

Point-to-Point Protocol

→ Point-to-point access

→ Byte-Oriented Protocol

4.14 Ethernet (IEEE 802.3)

4.14.1 Multiple Access Protocols

4.14.2 Pure ALOHA

A station may transmit a frame at any time. The station then listens for an amount of time equal to the maximum possible round-trip propagation delay ($2t_p$) for an acknowledgment.

If the station does not hear an acknowledgment, it resends the frame.

The maximum utilization of the channel is only about 18%.

4.14.3 Slotted ALOHA

Here, the channel is organized into uniform slots whose size equals the frame transmission time.

Transmission is permitted only at a slot boundary.

The maximum utilization of the system is about 37%.

Both ALOHA and slotted ALOHA fail to take advantage of one of the key properties of networks, which is that propagation delay; the stations may be very small compared to frame transmission time.

4.14.4 CSMA/CD (Carrier-Sense Multiple-Access with Collision Detection)

In CSMA/CD, the station listens to the medium while transmitting (Listen while talking).

The steps in CSMA/CD are:

(i) If the medium is idle, transmit.

(ii) If the medium is busy, continue to listen until the channel is idle, then transmit immediately (1-persistent).

(iii) If a collision is detected during transmission, transmit a jamming signal to ensure that all stations know that there has been a collision and then cease transmission.

(iv) After transmitting the jamming signal, wait a random amount of time, referred to as backoff, then attempt to transmit again.

4.14.5 The Binary Exponential Backoff Algorithm

This algorithm is used to calculate random amount of time (backoff) to wait before attempting to retransmit.

In this algorithm, after k-th collision, each station waits x-slot times where $0 \leq x \leq 2^k$.

We choose x randomly from the interval 0 to 2^{k-1}.

If R-stations transmit during a contention slot with probability p, the probability A that some station acquires the channel in that slot is:

$$A = Rp(1-p)^{R-1}$$

Frames should be long enough to allow collision detection prior to end of transmission.

Frame transmission time $(t_f) \geq 2t_p$

4.14.6 Limitation

Enforces rigid output pattern even when the traffic on the network is small.

Chapter 5

Routing Algorithms

5.1 Types of Routing Algorithms

(1) Non-adaptive / Fixed / Static Routing

(2) Flooding

(3) Random Routing

(4) Flow-based Routing

(5) Adaptive / Dynamic Routing

5.2 Fixed Routing or Non-adaptive Routing

A route is selected for each source-destination pair of nodes in the network.

The routes are fixed; they may only change if there is a change in the topology of the network.

Algorithms used are:

(i) Dijkstra's Algorithm

(ii) Bellman-Ford Algorithm

5.2.1 Advantages

→ Simple

→ Works well in a reliable network with stable load

→ Same for virtual circuit and datagram

5.2.2 Disadvantages

→ Lack of flexibility

→ Does not react to failure or network congestion

5.3 Flooding

→ Requires no network information.

→ Every incoming packet to a node is sent out on every outgoing line except the one it arrived on.

5.3.1 Measures to Improve Flooding

→ A hop counter – a hop counter may be contained in the packet header, which is decremented at each hop, with the packet being discarded when the counter becomes zero.

→ Use sequence numbers to avoid sending them out a second time.

5.4 Selective Flooding

The routers do not send every incoming packet out on every line, only on those lines that go approximately in the direction of the destination.

5.5 Flow-based Routing

Uses both topology and load information for routing.

Flow between a pair of nodes is relatively stable and predictable.

For a given line, if the capacity and average flow is known, it is possible to compute mean packet delay using queuing theory.

From the mean delays of all the lines, it is easy to calculate the flow-weighted average to get the mean delay for the whole subnet.

Then the routing algorithm finds the path of minimum average delay.

5.6 Adaptive Routing

Routing decisions change as conditions on the network change.

5.6.1 Two Principal Conditions That Change

(i) **Failure:** When a node fails.

(ii) **Congestion**

For adaptive routing to be possible, network state information must be exchanged among the nodes.

5.6.2 Algorithms Used

(i) Distance Vector Routing

(ii) Link-state Routing

5.7 Distance Vector Routing

5.7.1 Key Characteristics

→ Knowledge about the entire network is not necessary – only knowledge to neighbours.

→ Information sharing at regular interval.

→ Each node maintains a routing table having one entry for each node with two other fields: preferred next node and a cost estimate of the distance, based on one of the metrics.

Network Id	**Cost**	**Next Router**

This algorithm is also called as the Distributed Bellman-Ford algorithm and the Ford-Fulkerson algorithm.

Here each node receives the routing tables from all of its neighbours and updates its own routing table accordingly.

The main issue with Distance Vector Routing algorithm is that it reacts rapidly to good news but leisurely to bad news (count-to-infinity problem).

5.8 Link-State Versus Distance Vector Routing

→ Link-state routing converges more quickly.

→ Link-state routing requires more CPU power and memory than distance vector routing algorithm.

→ Link-state protocols are generally more scalable than distance vector protocols.

5.9 Hierarchical Routing

The routers are divided into regions, with each router knowing all the details about how to route packets to destinations within its own region but knowing nothing about the internal structure of other regions.

For large networks, we usually create multilevel hierarchy.

Chapter 6

Congestion Control

6.1 Common Causes of Congestion

→ As packets arrive at a node, they are stored in an input buffer. If packets arrive too fast, an incoming packet may find that there is no available buffer space.

→ Even very large buffers cannot prevent congestion, because of delay, timeouts, and retransmissions.

→ Slow processors may lead to congestion.

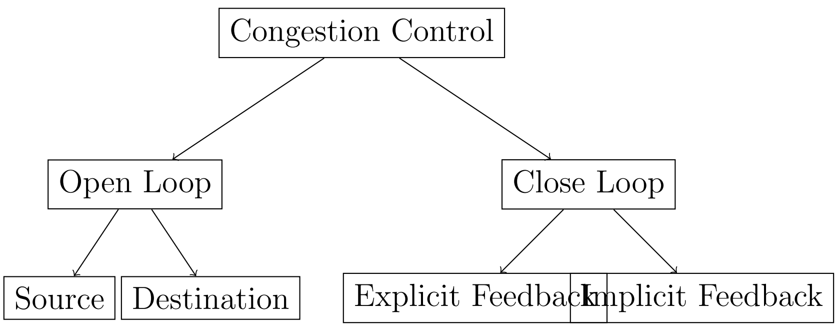
→ Low bandwidth line may also lead to congestion.

6.2 Congestion

When too many packets arrive in a part of the subnet, performance degrades. This situation is called congestion.

6.3 Congestion Control

Congestion Control refers to the mechanisms and techniques used to control congestion and keep the traffic below the capacity of the network.



6.4 Open Loop Congestion Control

The basic objective is to prevent congestion by adopting suitable policies for:

→ Flow Control

→ Acknowledgment

→ Retransmission (Timeout intervals)

→ Caching

→ Packet discard

→ Routing

6.4.1 Traffic Shaping

→ Leaky bucket algorithm

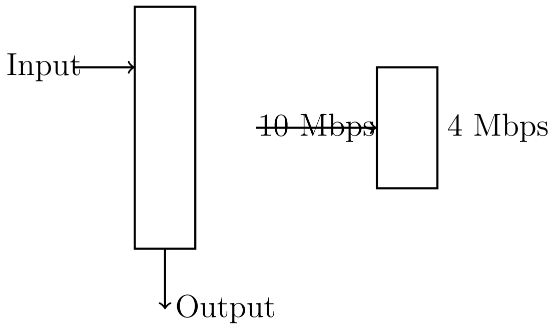
→ Token bucket algorithm

Traffic shaping is about regulating the average rate of data transmission.

6.5 The Leaky Bucket Algorithm

It shapes bursty traffic into fixed-rate traffic.

Packets are dropped when the bucket is full.



Limitation: Enforces rigid output pattern even when the traffic on the network is small.

6.6 Flow-based Routing

Uses both topology and load information for routing.

Flow between a pair of nodes is relatively stable and predictable.

For a given line, if the capacity and average flow is known, it is possible to compute mean packet delay using queuing theory.

From the mean delays of all the lines, it is easy to calculate the flow-weighted average to get the mean delay for the whole subnet.

Then the routing algorithm finds the path of minimum average delay.

6.7 Close Loop Congestion Control

6.7.1 Three Basic Steps

(1) Monitor the system for detection of congestion

(2) Pass the information to proper place for taking action to control congestion

(3) Adjust system parameters to get out of congestion

6.8 Congestion Control Versus Flow Control

Congestion Control	**Flow Control**
It is a global issue. It is a joint responsibility of the users and the network.	Flow control is local issue between a sender-receiver pair.
It occurs because of the combined behavior of the stations, switches, routing, etc.	Its job is to ensure that a fast sender does not overwhelm a slow receiver.
Its job is to ensure that the subnet is able to carry the offered load.	Its primary function is to ensure that a fast sender does not overwhelm a slow receiver.

6.9 Token Bucket Algorithm

In this algorithm, the leaky bucket holds tokens, generated by a clock at the rate of one token every ΔT sec.

For a packet to be transmitted, it must capture and destroy a token.

The token bucket algorithm allows idle hosts to save up permission to send large bursts later.

Chapter 7

Bridge

7.1 Network Devices Overview

→ Repeaters and hubs work in Physical Layer.

→ Bridge works in Data-Link Layer.

→ Router works in Network Layer.

→ Gateway works in Application Layer.

7.1.1 Repeaters

→ Connect different segments of a LAN.

→ It forwards every frame it receives.

→ It is a regenerator, not an amplifier.

7.1.2 Hub

→ Hub is a generic term but commonly refers to a multiport repeater.

→ It can be used to create multiple levels of hierarchy.

→ Easy to maintain and diagnose.

A bridge operates both in physical and data-link layers.

A bridge uses a table for filtering/routing.

It does not change the physical (MAC) addresses in a frame.

The difference between hub and bridge is that the hub forwards a frame from the input port to all other ports, while bridge forwards a frame only to the destination port.

It does not change the physical addresses in a frame.

7.2 Bridge Characteristics

7.2.1 Types of Bridges

(1) Transparent Bridges

(2) Source Routing Bridges

7.3 Transparent Bridge

7.3.1 Learning in Transparent Bridge

The stations are unaware of the presence of a transparent bridge.

Reconfiguration of the bridge is not necessary; it can be added/removed without being noticed.

7.3.2 Functions

(i) Forwarding of frames

(ii) Learning to create the forwarding table

7.3.3 Loop Problem in Transparent Bridges

Forwarding and learning processes work fine as long as there is no redundant bridge in the system.

On the other hand, redundancy is desirable from the viewpoint of reliability, so that the function of a failed bridge is taken over by a redundant bridge.

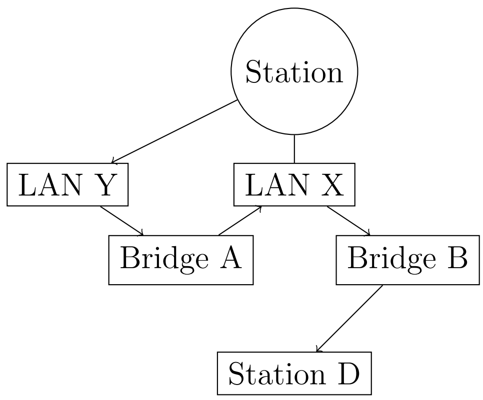
As redundancy creates a loop problem in the system, it is very undesirable.

7.3.4 Solution to Loop Problem (Spanning Tree Topology)

To solve the looping problem, the IEEE specification requires that the bridges use a special topology.

Such topology is known as spanning tree (a graph where there is no loop).

Without changing the physical topology, a logical topology is created that overlays the physical one.



If a frame is sent from station C, and if the destination of frame F is not present in the network, then the frame will start looping indefinitely between the bridges in the network.

7.4 Source Routing Bridge

7.4.1 Switch

→ Switching is a fast bridge.

→ Switch performs the forwarding operation using hardware, while bridges use software.

→ Ports are provided with buffer.

→ Each frame is forwarded after examining the address and forwarded to the proper port.

7.4.2 Three Forwarding Approaches

(i) Cut-through → No collision or error detection

(ii) Collision-free → No error detection

(iii) Fully buffered → Both collision and error detection

7.5 Router

Works in Network Layer.

It uses IP addresses.

A router has four basic components:

(i) Input Ports

(ii) Output Ports

(iii) Routing Processor

(iv) Switching Fabric

→ Input Port performs physical and data-link layer functions of the router.

→ Output Port performs the same functions as the input port, but in reverse order.

→ The routing processor performs the function of the network layer. The process involves the table lookup.

→ The switching fabric moves the packet from the input queue to the output queue by using specialized mechanisms.

Chapter 8

TCP/IP

8.1 TCP/IP

Acts as a glue to link different types of LAN and WAN to provide Internet; a single integrated network for seamless communication.

8.1.1 TCP/IP and OSI Model

<table>
<tr><td>APPLICATION</td><td rowspan="3">APPLICATIONS</td></tr>
<tr><td>PRESENTATION</td></tr>
<tr><td>SESSION</td></tr>
<tr><td>TRANSPORT</td><td>TCP / UDP</td></tr>
<tr><td>NETWORK</td><td>ICMP IP IGMP / ARP / RARP</td></tr>
<tr><td>DATA-LINK</td><td>ARPANET / LAN / SATNET / SONET</td></tr>
<tr><td>PHYSICAL</td><td>SLIP/PPP</td></tr>
</table>

(OSI) TCP/IP

8.2 Interworking Layers

(1) Addressing

(2) Packetizing

(3) Fragmentation and Reassembly

(4) Routing

(5) Flow Control

(6) Error detection / Error Control

(7) Congestion Control and QoS

(8) Security

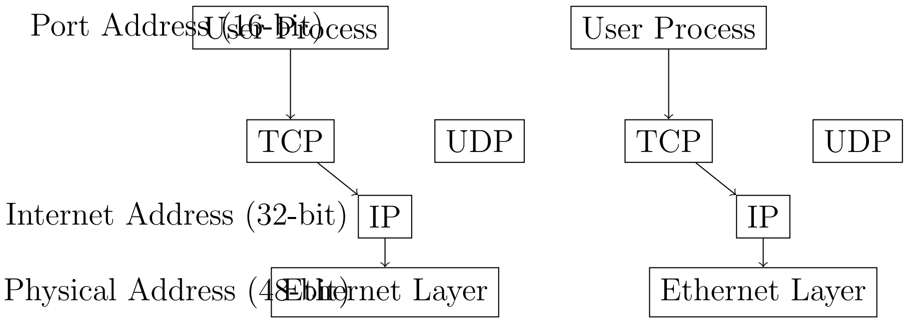
(9) Naming

8.2.1 IP

IP provides unreliable, connectionless best-effort datagram delivery service.

TCP provides reliable, efficient, and cost-effective end-to-end delivery of data.

8.3 Addresses in TCP/IP



8.4 Classful Addressing

Class A: 1.0.0.0 to 126.255.255.255

Class B: 128.0.0.0 to 191.255.255.255

Class C: 192.0.0.0 to 223.255.255.255

Class D: 224.0.0.0 to 239.255.255.255

Class E: 240.0.0.0 to 254.255.255.255

8.4.1 Default Masks for Classful Addressing

Class	**Binary**	**Dotted-Decimal**
A	11111111 00000000 00000000 00000000	255.0.0.0
B	11111111 11111111 00000000 00000000	255.255.0.0
C	11111111 11111111 11111111 00000000	255.255.255.0

8.4.2 Some Special IP Addresses

$\rightarrow$ 0.0.0.0 $\Rightarrow$ This host

$\rightarrow$ 255.255.255.255 $\Rightarrow$ Broadcast on this network (when netid is unknown)

first id **host id**

$00 - - - 00$ $\Rightarrow$ A host on this network

netid $111 - - - 111$

$\Rightarrow$ Broadcast on a distant network

$127 \cdot x \cdot y \cdot z$ $\Rightarrow$ Loopback

8.5 Subnetting

A subnet is a logically visible subdivision of an IP network.

The practice of dividing a network into two or more networks is called subnetting. Subnetting increases the number of 1s in the mask.

8.5.1 Classless Addressing

In this scheme, there are no classes, but the addresses are still granted in blocks.

8.5.2 Restrictions on Classless Address Blocks

(i) The addresses in a block must be contiguous.

(ii) The number of addresses in a block must be a power of 2 (1, 2, 4, —).

(iii) The first address must be evenly divisible by the number of addresses.

8.5.3 Example

Calculate first address and number of Addresses for 205.16.37.39/28.

Solution: $n = 28$, $(32 - n) = 4$

$(39)_{10} = (00100111)_2$

First Address ⇒

$$(00100000)_2 = (32)_{10}$$

⇒ 205.16.37.32

Last Address ⇒

$$(00101111)_2 = (47)_{10}$$

⇒ 205.16.37.47

Number of Addresses:

$$\Rightarrow 2^4 = 16 \text{ Ans}$$

8.5.4 Network Address

The first address in a block is normally not assigned to any device; it is used as the network address.

It is used as a YP network address.

Hierarchy:

8.5.5 Two-level Hierarchy: No Subnetting

Left-most bits	**Address X.Y.Z.t/n**	
n	defines the network	(32-n) right-most
bits define the particular host		

8.5.6 Three-levels of Hierarchy: Subnetting

Here, the rest of the world sees the organization as a single entity, however, internally, there are several subnets.

All messages are sent to the router address that connects the organization to the rest of the Internet.

The router routes the message to the appropriate subnets.

8.5.7 More Level of Hierarchy

There can be any number of hierarchical levels in classless addressing.

8.5.8 Example of Three Levels of Hierarchy

An organization is given the block 0/26, which contains 64 addresses. The organization has three offices and needs to divide the addresses into three subnets of 32, 16, and 16 addresses.

Solution:

(1) Suppose the mask for the first subnet is h_1, then $n_1 = 32 - h_1 = 27$

(2) Suppose the mask for the second subnet is h_2, then $2^{32-h_2} = 16 \Rightarrow h_2 = 28$

(3) Similarly for third subnet: $h_2 = 28$

This means we have the masks 27, 28, 28 with the organization mask being 26.

Address range for subnet 1:

17.12.14.0 to 17.12.14.31/27

Address range for subnet 2:

17.12.14.32 to 17.12.14.47/28

Address range for subnet 3:

17.12.14.48 to 17.12.14.63/28

8.6 Masking

A block of addresses can be defined as $x.y.z.t/n$ in which $x.y.z.t$ defines one of the addresses and the /n defines the mask.

A mask is a 32-bit number made of contiguous 1s followed by contiguous 0s.

The mask helps to find the netId and hostId.

Default masks for classful addressing:

Class A: $x.y.z.t/n$

in which $x.y.z.t$ defines one of the addresses and the /n defines the mask.

8.7 Calculating Addresses from Mask

First Address

Set the rightmost $(32 - n)$ bits to 1s.

Last Address

Set the rightmost $(32 - n)$ bits to 1s.

Number of Addresses

$$2^{(32-n)}$$

Example

Calculate first address and number of addresses for 205.16.37.39/28.

Solution: $n = 28$, $(32 - n) = 4$

$$(39)_{10} = (00100111)_2$$

First Address:

$$\begin{aligned}(00100000)_2 &= (32)_{10} \\ &\Rightarrow 205.16.37.32\end{aligned}$$

Last Address:

$$\begin{aligned}(00101111)_2 &= (47)_{10} \\ &\Rightarrow 205.16.37.47\end{aligned}$$

Number of Addresses:

$$2^4 = 16 \text{ addresses}$$

Chapter 9

Application Layer

9.1 TELNET

TELNET (Terminal Network) is a general-purpose client-server application program for remote log-in.

- Communication takes place using NVT (Network Virtual Terminal) character set.
- Telnet uses only one TCP connection.
- The server uses the well-known port 23 and the client uses an ephemeral port.
- To distinguish data from control characters, each sequence of control characters is preceded by a special control character called Interpret as Control (IAC).
- NVT uses two sets of characters, each 8 bits:
  - For Data (highest-order bit is 0)
  - For Control (highest-order bit is 1)

9.2 MIME (Multipurpose Internet Mail Extension)

MIME is a supplementary protocol that allows non-ASCII data to be sent through e-mail.
MIME defines 5 headers:

(i) MIME-Version – Current is 1.1

(ii) Content-Type – Content-Type/Content-Subtype

(iii) Content-Transfer-Encoding

(iv) Content-Id

(v) Content-Description

9.3 SMTP (Simple Mail Transfer Protocol)

SMTP simply defines four commands and responses must be sent back and forth.

- Uses Port 25 of destination machine.
- Commands are sent from the client to the server.
- It defines 14 commands:
  - HELO – Sender's host name
  - MAIL FROM – Sender of the message
  - RCPT TO – Intended recipient of message
  - DATA – Body of the mail
- Responses are sent from the server to the client. It is a 3-digit code.
- The process of transferring a mail message occurs in three phases:
  - Connection Establishment
  - Mail Transfer
  - Connection Termination

9.4 FTP (File Transfer Protocol)

FTP uses TCP/IP. It establishes two connections between the hosts:

- For Data transfer (Port 20)
- For Control Information (Port 21)

The control connection uses very simple rules of communication, while the data connection uses more complex rules due to the variety of data types transferred.

9.4.1 Communication over Control Connection

FTP uses the same approach as SMTP to communicate across the control connection. It uses the 7-bit ASCII character set.

9.5 POP3 (Post Office Protocol, version 3)

The client POP3 software is installed on the recipient's computer; the server POP3 software is installed on the mail server.

Uses TCP port 110 on the server.

Two modes of operation:

- Delete Mode
- Keep Mode

9.6 IMAP4 (Internet Mail Access Protocol version 4)

Uses TCP port 143.

IMAP4 provides the following extra functions:

(i) Checking email header prior to downloading.

(ii) Searching in the contents of the e-mail prior to downloading.

(iii) Partially download e-mail.

(iv) Create, delete, or rename mailboxes on the mail server.

(v) Creating a hierarchy of mailboxes in a folder for email storage.

The control connection remains connected during the entire FTP session. The data connection is opened and then closed for each file transferred.

9.7 DNS (Domain Name System)

DNS Lookup:

9.7.1 Reverse DNS Lookup

Determination of domain name from IP address.

9.7.2 Forward DNS Lookup

Determination of IP address from domain name.

9.8 HTTP Protocol

HTTP (HyperText Transfer Protocol) is a stateless protocol.

HTTP methods are case-sensitive, i.e., GET is a legal method but get is not.

HTTP 1.1 supports persistence connection. It is also possible to pipeline requests, i.e., sending request-2 before the response to request-1 has arrived.

HTTP supports proxy servers. A proxy server is a computer that keeps copies of responses to recent requests.

Chapter 10

Network Security

10.1 Public Key Cryptography

The public key used for encryption is different from the private key used for decryption.

10.1.1 RSA

Inventors: Rivest, Shamir, and Adleman.

Based on number theory.

The private key is a pair of numbers (d, n) and the public key is also a pair of numbers (e, n).

Steps:

(i) Choose two large primes p and q.

(ii) Compute:
$$n = p \times q \quad \text{and} \quad z = (p-1) \times (q-1)$$

(iii) Choose a number d, relatively prime to z.

(iv) Find e, such that:
$$(e \cdot d) \bmod z = 1$$

(v) For encryption:
$$C = P^e (\mathrm{mod} n)$$

(vi) For decryption:
$$P = C^d (\mathrm{mod} n)$$

10.1.2 Diffie-Hellman Key Exchange Algorithm

Given:

- Prime Number $= p = 7$
- Base $= g = 3$

Alice has a secret integer a, Bob has a secret integer b.
Secret Key $= g^{ab} \bmod p$
Example: Modulus = 7, Primitive Root = 3, $a = 2$, $b = 5$

$$\text{Key} = 3^{(2\times 5)} \bmod 7 = 3^{10} \bmod 7 = 4$$

At Alice:

- Primitive Root $= g$
- Prime Number = Modulus $= p$